

Homework 3

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Exercise 1

Let $m \in \mathbb{N}$ such that $-7 \equiv 6 \pmod{m}$. Then $(6 - -7) = 13 \equiv 0 \pmod{m}$. For this to be true, $m \mid 13$. Consequently $m = 1, 13$.

Exercise 2

Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{N}$ such that $a \equiv b \pmod{m}$. Then $a = b + mk$ for some $k \in \mathbb{Z}$. Let d such that $d \mid b$ and $d \mid m$. Then $d \mid b + mk$, and so $d \mid a$. Similarly, if we have some d such that $d \mid a$ and $d \mid m$, we can show that $d \mid b$. Since every common divisor of a and m is also a common divisor of b and m , the greatest common divisors must be the same, and so $(a, m) = (b, m)$.

Exercise 3

Let $a \in \mathbb{Z}$. Suppose a is odd. Then $a = 2k + 1$ for some $k \in \mathbb{Z}$ and so:

$$\begin{aligned}a^2 &= (2k + 1)^2 \\&= 4k^2 + 4k + 1 \\&= 4(k^2 + k) + 1\end{aligned}$$

Suppose k is odd. Then $k = 2l + 1$ for some $l \in \mathbb{Z}$ and so:

$$\begin{aligned}a^2 &= 4((2l + 1)^2 + (2l + 1)) + 1 \\&= 4(4l^2 + 4l + 1 + 2l + 1) + 1 \\&= 8(2l^2 + 3l + 1) + 1\end{aligned}$$

Now suppose k is even. Then $k = 2l$ for some $l \in \mathbb{Z}$ and so:

$$\begin{aligned}a^2 &= 4((2l)^2 + (2l)) + 1 \\&= 4(4l^2 + 2l) + 1 \\&= 8(2l^2 + l) + 1\end{aligned}$$

In either case, $a^2 = 8z + 1$ for some $z \in \mathbb{Z}$, and so $a^2 \equiv 1 \pmod{8}$.

Now suppose a is even. Then $a = 2k$ for some $k \in \mathbb{Z}$ and so:

$$\begin{aligned}a^2 &= (2k)^2 \\&= 4k^2 \\&= 4(k^2)\end{aligned}$$

Suppose k is odd. Then $k = 2l + 1$ for some $l \in \mathbb{Z}$ and so:

$$\begin{aligned}a^2 &= 4((2l + 1)^2) \\&= 4(4l^2 + 4l + 1) \\&= 8(2l^2 + 2l) + 4\end{aligned}$$

Now suppose k is even. Then $k = 2l$ for some $l \in \mathbb{Z}$ and so:

$$\begin{aligned}a^2 &= 4((2l)^2) \\&= 4(4l^2) \\&= 8(2l^2) + 0\end{aligned}$$

In either case, $a^2 \neq 8z + 1$ for some $z \in \mathbb{Z}$, and so $a^2 \not\equiv 1 \pmod{8}$. Consequently, a is odd iff $a^2 \equiv 1 \pmod{8}$.

Exercise 4

Part (a)

$$12x \equiv 16 \pmod{32}$$

Applying the division algorithm repeatedly:

$$32 = 12 \cdot 2 + 8$$

$$12 = 8 \cdot 1 + 4$$

$$8 = 4 \cdot 2 + 0$$

Since $(12, 32) = 4$ and $4 \mid 16$, a solution does exist. We then solve the equivalent problem:

$$3x \equiv 4 \pmod{8}$$

We now find the inverse of 3 modulo 8. Applying the division algorithm repeatedly:

$$8 = 3 \cdot 2 + 2$$

$$3 = 2 \cdot 1 + 1$$

$$2 = 1 \cdot 2 + 0$$

Consequently, $(34709, 100313) = 1$, so the inverse exists. Expressing the GCD as a linear combination:

$$\begin{aligned} 1 &= 2 \cdot -1 + 3 \\ &= (3 \cdot -2 + 8) \cdot -1 + 3 = 3 \cdot 3 + 8 \cdot -1 \end{aligned}$$

The inverse of 3 modulo 8 is then 3. Solving the congruence:

$$3 \cdot 3x \equiv 3 \cdot 4 \pmod{8}$$

$$x \equiv 4 \pmod{8}$$

Part (b)

$$481x \equiv 627 \pmod{703}$$

Applying the division algorithm repeatedly:

$$703 = 481 \cdot 1 + 222$$

$$481 = 222 \cdot 2 + 37$$

$$222 = 37 \cdot 6 + 0$$

Since $(481, 703) = 37$ and $37 \nmid 627$, no solutions exist.

Exercise 5

Part (a)

$$4^{-1} \bmod 17$$

Applying the division algorithm repeatedly:

$$17 = 4 \cdot 4 + 1$$

$$4 = 1 \cdot 4 + 0$$

Consequently, $(17, 4) = 1$, so the inverse exists. Expressing the GCD as a linear combination:

$$1 = 4 \cdot -4 + 17$$

The inverse of 4 modulo 17 is then -4 , or equivalently, 13.

Part (b)

$$8^{-1} \bmod 35$$

Applying the division algorithm repeatedly:

$$35 = 8 \cdot 4 + 3$$

$$8 = 3 \cdot 2 + 2$$

$$3 = 2 \cdot 1 + 1$$

$$2 = 1 \cdot 2 + 0$$

Consequently, $(35, 8) = 1$, so the inverse exists. Expressing the GCD as a linear combination:

$$1 = 2 \cdot -1 + 3$$

$$= (3 \cdot -2 + 8) \cdot -1 + 3 = 3 \cdot 3 + 8 \cdot -1$$

$$= (8 \cdot -4 + 35) \cdot 3 + 8 \cdot -1 = 8 \cdot -13 + 35 \cdot 3$$

The inverse of 8 modulo 35 is then -13 , or equivalently, 22.